Direct variation

1 If $(4, 512)$ is on the curve, $P = kQ^4$, solve for k.

2 A variable M is directly proportional to the square of N.

a Using k as the constant of proportionality, form an equation for M in terms of N.

b Given that $(4, 64)$ satisfies the equation, solve for k.

3 Consider the relationship where y is directly proportional to the cube of x.

a Using k as a constant of proportionality, state the equation relating x and y.

b The following table of values shows the relationship between x and y:

x	1	2	3	4	5
y	3	24	81	192	375

Solve for k, the constant of proportionality.

4 y is a power function which is directly proportional to x^n.

a Using k as the proportionality constant, form an equation for y in terms of x.

b The table below shows values that satisfy the relationship $y = kx^n$, for the case when $n = 2$.

x	2	4	6	8	10
y	32	128	288	512	800

Solve for the constant of proportionality k. Round your answer to the nearest integer.

5 y is a power function which is directly proportional to x^n.

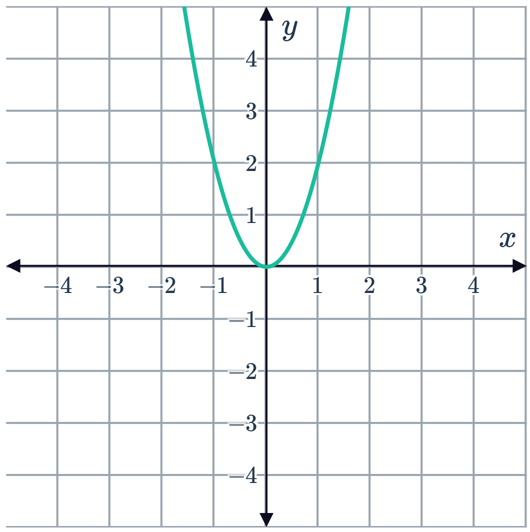
a Using k as the proportionality constant, form an equation for y in terms of x.

b The table below shows values that satisfy the relationship $y = kx^n$, for the case when $n = 3$.

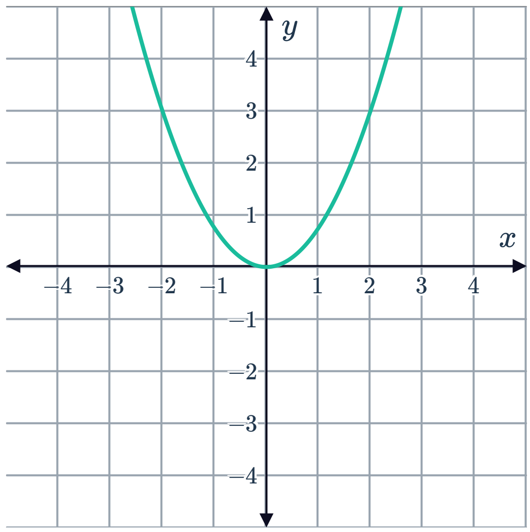
x	2	4	6	8	10
y	-48	-384	-1296	-3072	-6000

Solve for the constant of proportionality k. Round your answer to the nearest integer.

6 For the quadratic function pictured here, determine the constant of proportionality k:



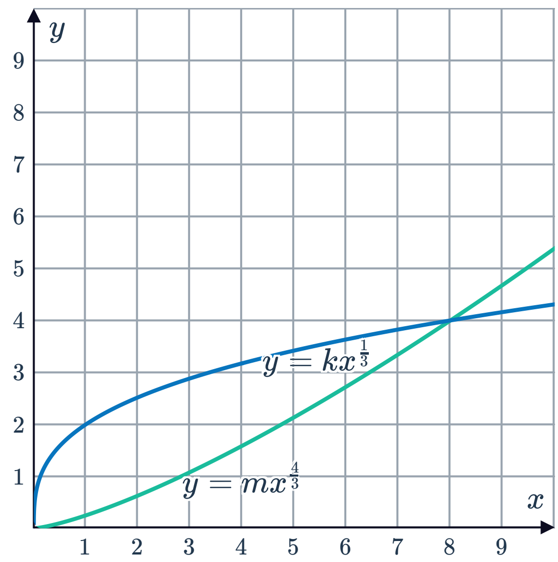
7 For the quadratic shown, determine the constant of proportionality k:



8 The following graph shows the intersection of the two functions $y = mx^{\frac{4}{3}}$ and $y = kx^{\frac{1}{3}}$:

a Find the value of m.

b Find the value of k.



Inverse variation

9 If y varies inversely with x, write an equation that uses k as the constant of variation.

10 If $(10, 15)$ is on the curve $P = \frac{k}{Q}$, solve for k.

11 Determine whether each of the following is an example of a direct variation or an inverse variation:

a The variation relating the distance between two locations on a map and the actual distance between the two locations.

b The variation relating the number of workers hired to build a house and the time required to build the house.

12 If r varies inversely with a, write an equation that uses k as the constant of variation.

13 Determine whether the following equations represent an inverse relationship between x and y:

a $x = 1 + y^3$

b $x = \frac{8}{y^2}$

c $y = 6x + 8$

d $xy = -7$

e $x = \frac{2}{y}$

f $xy = 5x$

14 Consider the equation $s = \frac{375}{t}$.

a State the constant of proportionality.

b Find the exact value of s when $t = 6$.

c Find the exact value of s when $t = 12$.

15 State whether the following tables represent an inversely proportional relationship between x and y.

a

x	1	2	3	4
y	3	1.5	1	0.75

b

x	1	2	3	4
y	36	18	12	9

c

x	1	5	6	10
y	3	75	108	300

d

x	1	2	3	4
y	4	5	6	7

16 m is proportional to $\frac{1}{p}$. Consider the values in the table which represents the relationship.

p	4	6	7	x
m	63	y	36	28

a Determine the constant of proportionality, k.

b Find the values of x and y.

17 Find the equation relating t and s for each of the following tables of values:

a

s	1	2	3	4
t	48	24	16	12

b

s	3	6	9	12
t	$\frac{2}{9}$	$\frac{1}{18}$	$\frac{2}{81}$	$\frac{1}{72}$

18 Use technology to graph the inverse relationship $y = \frac{6}{x}$.

a How many x-intercepts does the graph have?

b How many y-intercepts does the graph have?

Applications

19 In exchange rate calculations the constant of proportionality is the exchange rate itself. Determine the exchange rate, k, if someone can purchase:

a 889.00 AUD with 700.00 USD.

b 553.00 USD with 700.00 AUD.

20 The heart mass H of a mammal is proportional to its body mass M.

a Using k as the constant of proportionality, form an equation relating H and M.

b Given that a human with a body mass of 77kg has a heart mass of 0.462kg, determine k, the constant of proportionality.

c If a Chimpanzee has a body mass of 57kg, what is the heart mass H?

d If a Macaque has a heart mass of 0.084kg, solve for its body mass M.

21 The blood circulation time C of a mammal is proportional to the fourth root of its body mass B.

a Using k as the constant of proportionality, form an equation relating C and B.

b Given that an elephant with a body mass of $5290 \mathrm{~kg}$ has a blood circulation time of 148 seconds, determine k, the constant of proportionality. Round your answer to one decimal place.

c If a Orangutang has a body mass of $40 \mathrm{~kg}$, find C, its circulation time. Round your answer to one decimal place.

d If a Squirrel has a circulation time of 22 seconds, what is the body mass? Round your answer to one decimal place.